Krishin Lal

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Professional Summary

AI Engineer & Data Scientist with hands-on production experience building LLM-powered systems, agentic pipelines, OCR automation, geospatial analytics, and deep learning forecasting at one of Pakistan's largest banks. Specializes in end-to-end AI system design, from embedding-based entity matching and vector retrieval to FastAPI deployment on Azure ML, with a consistent record of replacing manual workflows with scalable, measurable AI solutions.

Experience

Data Scientist & AI Engineer, United Bank Limited (UBL) – Karachi, Pakistan July 2025 – Present

- **Compliance Chatbot:** Built a RAG advisory chatbot over 300+ GB of regulatory and government documents from 15 bodies. Parent-child chunking, a ∼35-field metadata layer, and hybrid sparse + dense search keep retrieval accurate and no information is lost between chunks.
- **Audit Automation Suite:** Built two LLM tools: an RCM generator that creates Risk Control Matrices from regulatory circulars (unlimited files), and an audit report reviewer that corrects grammar, terminology, and formatting with color-coded highlights and a full change log. Together they save 8+ hours of manual work daily.
- **Procurement Invoice OCR:** Production multi-class invoice pipeline (Azure Document Intelligence, GPT-4o, FastAPI) that cut processing from 5 to 1 minute per invoice (80%), reclaiming 200+ hours monthly across 150 invoices/day.
- **Trade Document OCR:** Replaced a 5-hour daily manual process with a 15-minute pipeline, with a custom LLM validation layer that handles handwritten cover pages, saving 1,467 hours/year (1,500 to 33).
- **Account Opening Automation:** End-to-end OCR pipeline for account opening forms from all UBL branches, extracting KYC fields and signatures into validated JSON. Eliminated a costly third-party vendor and created a reusable in-house document-AI framework.
- **NTB/ETB Entity Matching Engine:** Multi-stage matching (exact, containment, embedding) using SentenceTransformers and cosine similarity to reconcile thousands of CRM company names against bank records, classifying 2,479 companies as Existing- or New-to-Bank prospects for business development.
- **Geospatial Branch Expansion Planning:** Mapped 2,200+ UBL branches and geocoded 8,000+ competitor locations across five major cities. Built 1×1 km grids with a 500 m cannibalization buffer to produce 116 data-backed branch opportunity zones that informed executive expansion decisions.

Projects

OmniVault - Privacy-First Bookmarking App | Android, Kotlin

- Built a fully local Android app that unifies saved content from Instagram, X, YouTube, LinkedIn, and the web in one place. Tapping Share auto-fetches the platform, title, caption, description, and thumbnail, then sorts each saved item (a "stream") by platform and content type.
- Added a distraction-free, ad-stripped reader, quick notes, custom categories, priorities, and deadlines with reminder notifications. No accounts and no tracking: nothing leaves the device.

Multi-Agent Narrative System | Hackfest x Datathon – Generative AI Competition Feb 2026

- Built a LangGraph + Google Gemini system where a Director agent orchestrates multiple character agents, each with persistent memory and a distinct personality.
- Added a non-verbal action system and memory-sharing protocols so characters express emotion and retain context across long story arcs.

EDNOVA - Gamified SaaS Learning Management System (Final Year Project) Mar 2024 – Jun 2025

- Built a full-stack multi-tenant LMS (MERN) with role-based access, real-time engagement tracking, a points-based reward engine, and leaderboards.
- Created an analytics dashboard with composite engagement scores that helps educators identify at-risk students early.

Skills

AI & LLM Engineering: LangChain, LangGraph, LLMs, RAG Architecture, Agentic Systems, Multi-Agent Orchestration, Prompt Engineering, Semantic Search, Vector Embeddings, Azure OpenAI

Computer Vision & OCR: Azure Document Intelligence, Document Layout Analysis, Table Extraction, Handwriting Recognition, Image Preprocessing

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, XGBoost, N-BEATS, NHITS, SARIMA, ARIMA, DeepAR, LSTM, Prophet, NeuralForecast, Time-Series Forecasting, Hyperparameter Tuning

NLP & Embeddings: SentenceTransformers, HuggingFace, Cosine Similarity, Entity Matching, Semantic Blocking, ChromaDB, Vector Databases

Geospatial Engineering: SERP API, Coordinate Extraction, Grid-based Spatial Analysis, Blind Spot Detection, Cannibalization Radius Modeling, Opportunity Zone Mapping

Deployment & MLOps: FastAPI, Azure Machine Learning, Docker, Model Containerization, REST API Design, API Authentication, Endpoint Monitoring

Data Engineering & Databases: Pandas, NumPy, SQL, Teradata, MongoDB, SQL Server, Feature Engineering, EDA

Mobile Development: Kotlin, Android

Tools & Platforms: Python, SQL, JavaScript (MERN), Git, GitHub, Docker, Jupyter, VS Code, Azure Cloud, Kubeflow

Education

Iqra University, Karachi – Bachelor of Science in Computer Science Oct 2021 – Jun 2025
Government Degree College, Kandhkot – Intermediate (Science) 2019 – 2021

Key Achievements

- 1,467 hours/year saved through trade document OCR automation (1,500 to 33 hours/year).
- 200+ hours/month reclaimed via invoice automation, 4 minutes saved per invoice across 150 daily invoices.
- 116 branch opportunity zones identified from 8,000+ competitor locations, shaping UBL's Annual Branch Expansion Plan.
- 2,479 companies classified as NTB/ETB prospects through semantic embedding matching.
- 600+ daily inference requests served across production ML endpoints on Azure Machine Learning.
- Eliminated third-party vendor dependency with in-house account opening OCR automation.